

RESEARCH ARTICLE SUMMARY

SOLAR CELLS

Bypassing the yellow phase for extremely stable formamidinium lead iodide perovskite solar cells

Rabindranath Garai, Isaac Metcalf, Nilanjana Nandi, Paramvir Ahlawat, Braulio Reyes-Suárez, Faiz Mandani, Hao Zhang, Tim Kodalle, Michael D. Irwin, Claudine Katan, Carolin M. Sutter-Fella, Mercouri G. Kanatzidis, G. N. Manjunatha Reddy, Jacky Even, Aditya D. Mohite*



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INTRODUCTION: Formamidinium lead iodide (FAPbI₃) emerged as an ideal material for single-junction perovskite solar cells owing to its near optimal bandgap of 1.45 to 1.5 eV and outstanding thermal stability. However, the photoactive cubic α -phase (3C-FAPbI₃) of FAPbI₃ is structurally unstable and undergoes a reconstructive phase transition to the nonperovskite yellow hexagonal δ -phase (2H-FAPbI₃) at ambient temperature. The phase reconstruction from 3C-FAPbI₃ to 2H-FAPbI₃ could be prevented by alloying methylammonium (MA) or Cs or both at the A-site and Br at the halide site, but this limits long-term durability owing to phase segregation or materials instability. Addressing these challenges requires a rational design strategy to stabilize 3C-FAPbI₃ by restricting lattice reconstruction without compromising thermal stability.

RATIONALE: Two main strategies have emerged to improve the phase stability and film quality of FAPbI₃. The first is a lattice-templating approach, which enables the slow formation of perovskite but it eventually degrades through the formation of yellow phases. The second approach, which has been widely explored, involves additive engineering, using alkyl ammonium halide or mostly chloride-based additives, which provide better control over the crystallization route. However, the FAPbI₃ films fabricated using these additives are often alloyed and compromise long-term stability. Moreover, the exact role of Cl has been unclear and speculative, specifically when Cl-based additives are used. Even after using a high additive concentration, the incorporation of Cl in perovskite lattice is rare.

RESULTS: Guided by synergistic modeling and experimental studies, we developed a coadditive strategy using 15 mol % FAPbI₃

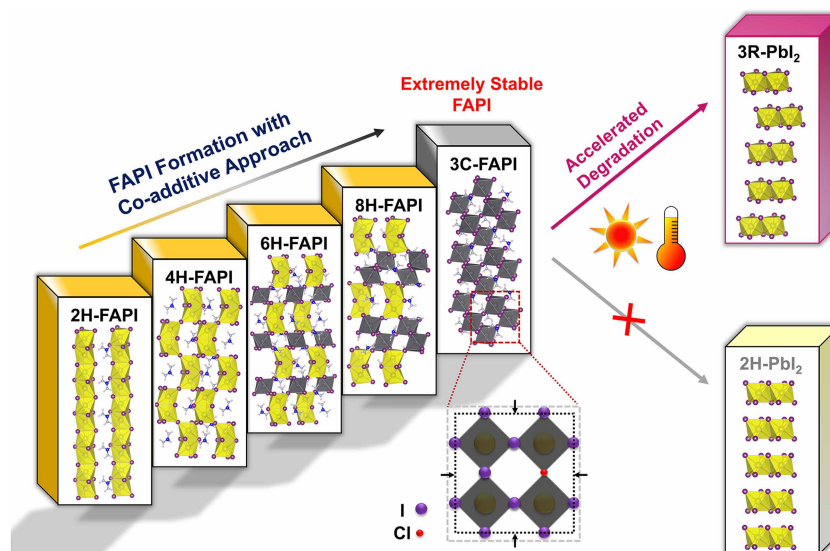
and 0.5 mol % BA₂PbI₄ perovskites in combination (where BA is butylammonium) to enable a highly oriented (100) Cl-doped FAPbI₃ film with exceptional durability. Synchrotron-based in situ wide-angle x-ray scattering revealed a favorable transition for the coadditive-treated FAPbI₃ (FAPbI₃-CA) to the corner-sharing 3C black phase through a progressive transformation through the 2H, 4H, 6H, and 8H phases. Moreover, solid-state ³⁵Cl nuclear magnetic resonance (NMR) revealed Cl incorporation in the perovskite lattice and, as predicted by modeling, indicated that Cl plays a key role in altering the energetics of both the formation and degradation pathways. The Cl-doped perovskite can completely bypass the expected and energetically favorable degradation pathway via the yellow phase or the 2H-PbI₂ phase. Instead, it undergoes degradation only upon exposure to harsh conditions such as 15-sun illumination and 90°C through the energetically uphill 3R-PbI₂ phase path. A p-i-n device fabricated with FAPbI₃-CA film demonstrated a power conversion efficiency (PCE) of 25.1% with an average of 24.1% (40 devices). The notable film stability translated to other devices and retained 98% of its initial PCE under open-circuit conditions at 85° ± 5°C for 1200 hours.

CONCLUSION: Our study highlights the decisive role of chloride in regulating both the formation and degradation pathways. This regulation is critical for creating a perovskite film with commercially relevant durability. □

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The formation and degradation pathways for extremely stable Cl-doped FAPbI₃

FAPbI₃ fabricated with a coadditive approach using FAPbI₃ and BA₂PbI₄ perovskites demonstrated a favorable phase transition from face-sharing 2H-4H-6H-8H to the corner-sharing 3C black phase. FAPbI₃ incorporated Cl into the system, and BA₂PbI₄ perovskites improved crystallinity; both additives synergistically induced a comprehensive lattice strain to enable outstanding stability.



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Bypassing the yellow phase for extremely stable formamidinium lead iodide perovskite solar cells

Rabindranath Garai¹, Isaac Metcalf¹, Nilanjana Nandi¹, Paramvir Ahlawat², Braulio Reyes-Suárez³, Faiz Mandani¹, Hao Zhang¹, Tim Kodalle^{4,5}, Michael D. Irwin⁶, Claudine Katan⁷, Carolin M. Sutter-Fella⁵, Mercouri G. Kanatzidis^{8,9}, G. N. Manjunatha Reddy³, Jacky Even¹⁰, Aditya D. Mohite^{1,11*}

Using modeling and structural studies, we show that chloride incorporation in formamidinium lead iodide (FAPbI₃) perovskites alters the energetics of both the formation and degradation pathways. We fabricated films with two coadditives [15 mole % FA chloride (FACl) and 0.5 mole % BA₂PbI₄, where BA is butylammonium], in which FACl ensures chloride incorporation and both additives collectively create a compressive lattice strain that stabilizes the FAPbI₃ black phase and bypasses the formation of a yellow phase during degradation. The coadditive strategy revealed a favorable transition from face-sharing 2H, 4H, 6H, and 8H phases to the corner-sharing 3C black phase. Photovoltaic devices with a p-i-n architecture had an average power conversion efficiency (40 devices) of 24.1% and lost only 2% of their efficiency after 1200 hours at 85° ± 5°C, 1-sun illumination, and open-circuit conditions. Upon stressing at 15-sun illumination at 90°C for >400 hours, the stabilized black 3C phase degraded not through the 2H-PbI₂ phase but through the energetically uphill 3R-PbI₂ phase.

Perovskite films based on formamidinium (FA⁺), such as FAPbI₃ (or FAPI), exhibit near optimal bandgaps (E_g) of 1.45 to 1.5 eV and demonstrate outstanding thermal stability in comparison to methylammonium films such as MAPbI₃ (MAPI, where MA is methylammonium) or mixed-cation systems (1–4). However, the high formation energy of three-dimensional (3D) bulk FAPI makes the black photoactive cubic α -phase (also known as 3C-FAPI) thermodynamically unstable, which usually undergoes a reconstructive phase transition to the photoinactive yellow hexagonal δ -phase (also known as 2H-FAPI) at ambient temperature. Realizing pure FAPI perovskites with no mixed A-site cation or halide alloying has been challenging because of their inherent phase instability and slow crystallization kinetics, which can further compromise the film crystallinity and stability (5–7). Alloying MA or Cs or both at the A-site and Br at the halide site makes the reconstruction to the hexagonal phase less favorable. However, this approach compromises long-term durability by increasing the propensity for phase segregation under light, temperature, humidity, or combinations thereof (8–11). Moreover, such alloying also increases the bandgap.

Two main strategies have emerged for the stabilization of the FAPI perovskite phase and film quality. The first is a lattice-templating approach, which enables the slow formation (over several minutes) of 3C-FAPI (black phase) from 2H-FAPI, but without the formation of any intermediate polytype phases during crystallization. Doherty *et al.*

showed that surface-bound ethylenediaminetetraacetic acid acted as a molecular template, which led to octahedral tilting and stabilized the FAPI phase (12). More recently, templating with carefully chosen lattice-matched $n = 2$ FA-based two-dimensional (2D) perovskites induced a compressive lattice strain in FAPI films after formation to avoid reconstruction to the yellow phase (6, 13, 14). The 2D perovskite-stabilized FAPI yielded photovoltaic devices with excellent durability under 85°C at maximum power point tracking (MPPT). However, this FAPI film ultimately degrades through the formation of the 2H phase of PbI₂.

The second approach, additive engineering, uses alkyl ammonium halide or pseudohalide salts such as methylammonium chloride (MACl), dimethylammonium chloride (DMACl), methylammonium thiocyanate, methylene diammonium dichloride, formamidinium formate, propyl ammonium, and isopropyl ammonium chloride (15–20). These salts (mostly chloride additives) induce intermediate-phase formation and provide faster crystallization kinetics. McMeekin *et al.* reported the formation of stable mixed iodine-bromine perovskites through an intermediate-phase engineering process with DMACl (21). The addition of alkyl ammonium chlorides can induce crystallization pathways with mixed edge and corner-sharing FAPI intermediates (4H, 6H) during 2H-FAPI-to-3C-FAPI lattice reconstruction (18–20). Marchezi *et al.* proposed that the nucleation of these FAPI polytypes can be regulated by varying the alkyl chain length of alkyl ammonium chlorides (22). Recently, subzero-temperature processing (solution, antisolvent, substrate, and processing atmosphere all at –20°C) of FAPI with MACl additive enabled the formation of highly crystalline intermediate phases and perovskite films with reduced defect density (23).

We hypothesized that the stepwise growth of FAPI through the hexagonal polytypes as intermediates presents a superior crystallization pathway and eliminates nanoscale delta-phase impurities. However, FAPI films fabricated with additives are usually alloyed with MA at the A-site or with other long-chain amines that often form heterogeneous quasi-2D perovskite films. As a result, accelerated aging tests under constant heating at 85°C and solar illumination either have not been performed or have resulted in compromised stability. In addition, the exact role of chloride has been unclear, specifically when Cl-based additives are used. It is usually assumed that the chloride leaves as hydrochloride (HCl) gas during crystallization, and, to date, there has been no direct evidence for the presence of chloride ions in the FAPI lattice. Finally, very few studies have investigated the accelerated degradation pathways in stabilized FAPI perovskite films (24).

We now demonstrate through a synergistic, modeling-guided synthesis approach that a Cl-doped FAPI film, stabilized using FACl and 2D perovskites (a 15 mol % FACl and 0.5 mol % BA₂PbI₄ combination, where BA is butylammonium) results in a highly oriented (100) film with exceptional durability. Comprehensive time-resolved solid-state ³⁵Cl nuclear magnetic resonance (NMR) as a function of annealing temperature confirmed the presence of Cl in the FAPI lattice. Our modeling studies suggested that the Cl in the lattice altered the energetics of both the formation and degradation processes. In situ film formation with x-ray scattering revealed an energetically favorable transformation from the face-sharing 2H, 4H, 6H, and 8H FAPI phases into a corner-sharing 3C black phase. The film did not undergo degradation under 1-sun illumination and 85° ± 5°C (equivalent to being measured under open-circuit conditions and thermal stress) but underwent degradation under concentrated solar illumination of 15 suns and 90°C.

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Accelerated aging at 15-sun illumination and elevated temperatures of 65°, 75°, and 90°C for >400 hours showed that the film did not degrade through the energetically favorable yellow hexagonal δ -phases, as previously observed (10, 25), but instead through the thermodynamically uphill 3R phase of lead iodide. We attribute the formation of 3R-PbI₂ to a compressive strain caused by Cl doping, which exhibited a d-spacing closer to the (104) plane of 3R-PbI₂, thereby facilitating the transition. The excellent stability of the coadditive-stabilized FAPI films was translated to efficient photovoltaic devices with a p-i-n architecture with a champion power conversion efficiency (PCE) of 25.1% (average PCE 24.1% out of 40 devices). The devices lost 2% of their PCE (T_{98}) after 1200 hours under 1-sun illumination, 85° ± 5°C, and open-circuit voltage (V_{OC}) conditions. Our results highlight the decisive role of chloride in regulating both the formation and degradation pathways, which is critical for creating a perovskite film with commercially relevant durability.

Prediction of formation and degradation of FAPI through polytypes

We calculated the detailed formation pathway of FAPI using density functional theory (DFT), both with and without chloride additive (Fig. 1A). The phase transformation from a 100% face-sharing hexagonal polytype 2H to a fully corner-sharing cubic polytype 3C often requires high energy. FAPI crystallization typically follows this pathway in the absence of any other halides. However, bromide additives can facilitate the formation of intermediate hexagonal perovskitoid polytypes such as 4H, 6H, and 8H and provide a thermodynamically favorable phase transition. Gratia *et al.* identified a crystallization sequence for (FAPbI₃)_x(MAPbBr₃)_{1-x} involving 2H, 4H, 6H, and 3C polytypes, and later studies revealed the emergence of higher-order polytypes (8H, 10H, 12H) and stack-faulted structures (26–30).

These polytypes share a mixed corner-sharing and face-sharing octahedral connectivity but show different stacking sequences, where the corner-sharing component gradually increases from the 4H to 8H phases, thus providing an energetically favorable transition pathway through a multistep growth mechanism. However, the I-Br mixed system undergoes halide segregation under illumination. By contrast, chloride incorporation in an iodide-based perovskite is generally thought to leave as HCl gas after annealing, thereby offering nearly phase-pure perovskite films. As discussed in the next section, our NMR experiments suggest that chlorine-based additives can incorporate Cl into the FAPbI₃ lattice without any observable halide segregation over time.

To understand the effect of iodine substitution by chlorine, we first conducted DFT comparative calculations in pure 2H, 4H, 6H, 8H, 10H, 12H, and 3C polytypes, including van der Waals-corrected functionals [b86bpbe+xDM

(31) and PBE+D3] within the Quantum ESPRESSO framework (see supplementary methods for more details). We assessed the relative energies of FAPbI₃ and FAPbCl₃ polytypes as a function of increasing corner-sharing octahedral content (Fig. 1B and fig. S1A). Chloride substitution energetically favored corner-sharing arrangements in higher-order polytypes over the fully face-sharing 2H structure.

To further capture finite temperature and size effects, we also used large-scale molecular dynamics simulations on the same polytype structures using the GRACE (32) machine-learning interatomic potential, trained on extensive DFT datasets. Conducted on 5000-atom supercells at atmospheric conditions, these simulations spanned 2H to 3C polytypes with varied Pb-X (X = I or Cl) connectivity (fig. S1B). A substantial energy reduction that stabilized corner-sharing polytypes relative to the less favorable face-sharing 2H structure underscored a thermodynamic preference for corner-sharing motifs in chloride-substituted systems. Our study suggests that the face-sharing octahedra could be

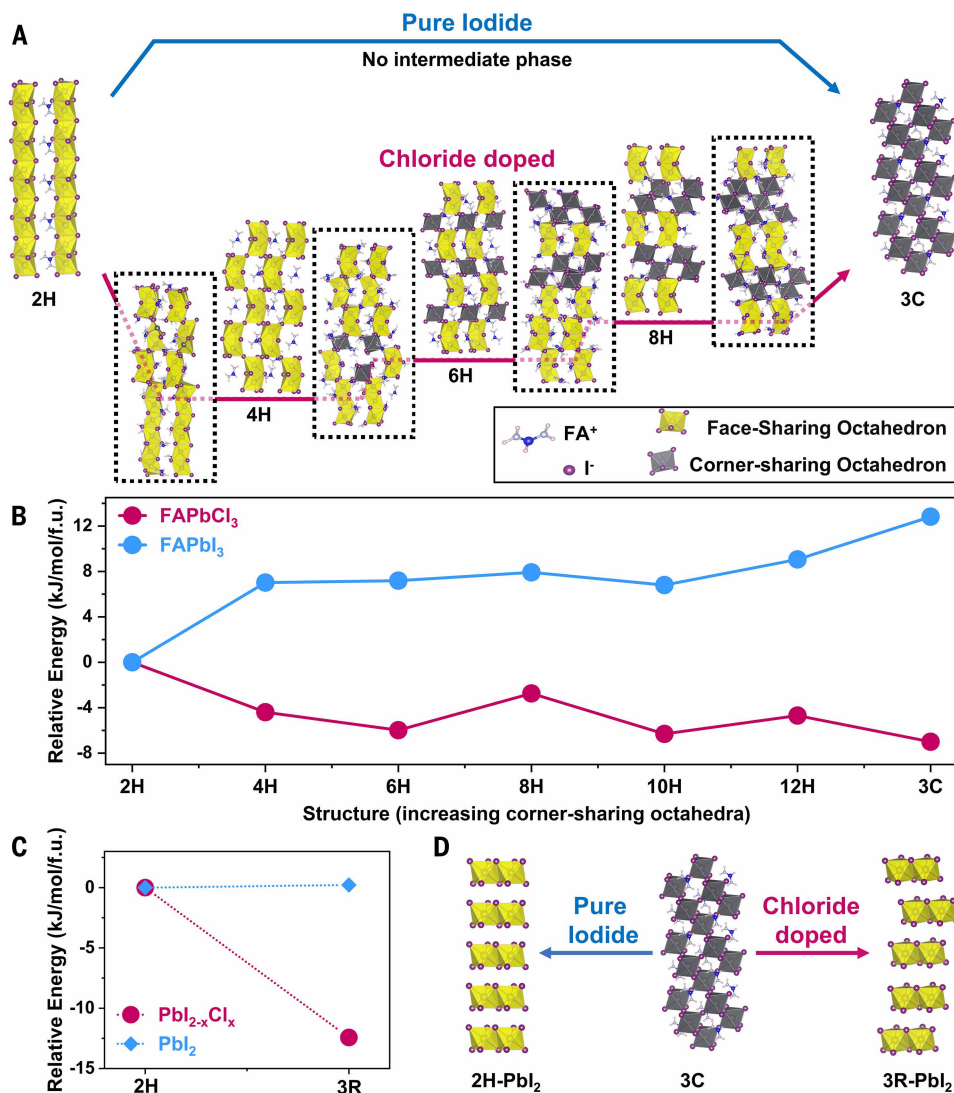


Fig. 1. Calculated formation and degradation of FAPI through polytypes. (A) Formation pathway of FAPI with and without chloride. (B) The relative energies of the FAPI polymorphs, calculated using DFT with the xDM van der Waals corrections. Energies are plotted as a function of the corner-sharing octahedral content, illustrating the effect of chloride substitution on the relative stability of different polytypes (e.g., 2H to 3C) as illustrated in (A). f.u., formula units. (C) The relative energies of the PbI₂ and PbI_{2-x}Cl_x polymorphs, calculated using DFT with the xDM van der Waals correction (dotted lines are a guide to the eye). (D) Degradation pathway of 3C-FAPI obtained with and without chloride additive, showing that 3C-FAPI finally decomposes to the 3R and 2H phases of PbI₂, respectively.

eliminated in the final film with chlorine additives and lead to the formation of a stable black-phase FAPI.

The degradation pathway was also expected to be distinct with and without the chloride additive. We conducted further DFT calculations (b86bpbexDM and PBE+D3) on PbI_2 , a major degradation product of FAPI. We assessed the relative energies of the different phases of $\text{PbI}_{2-x}\text{Cl}_x$ (2H- PbI_2 and 3R- PbI_2). The 2H- PbI_2 phase was energetically more favorable for only the iodide system. Replacing the iodide with chloride in the same structures allowed formation of energetically preferred 3R- $\text{PbI}_{2-x}\text{Cl}_x$ (Fig. 1C and fig. S2). The edge-sharing octahedron layers were placed exactly one above another in the 2H- PbI_2 phase, whereas the layers were slightly shifted in their stacking in the 3R phase. This structure was evidenced by one-dimensional x-ray diffraction (XRD) analysis (fig. S3). The probable degradation pathway from 3C-FAPI to different phases of PbI_2 would be strongly dependent on the extent of chloride doping (Fig. 1D). Therefore, these results collectively indicate that a small amount of Cl doping in the FAPI system should provide an energetically favorable formation pathway that leads to a different degradation pathway.

In situ FAPI formation and film characterization

We performed several in situ formation and film characterization studies with several additives with and without chlorine. We show that the coadditive-treated FAPI film (a 15 mol % FACl and 0.5 mol % BA_2PbI_4 combination), or FAPI-CA film, exhibited a superior crystallization pathway through all the polytypes and resulted in high-quality Cl-doped FAPI film.

To test the results from our modeling studies, we performed different in situ experiments with various chloride additives, such as MACl, FACl, and PbCl_2 , to regulate the crystallization kinetics of FAPI. We selected these additives because MACl is a state-of-the-art stabilizer for FAPI, and FACl and PbCl_2 do not introduce any additional cations. For comparison, we also studied a Cl-free 2D-templated FAPI system in which BA_2PbI_4 crystal seeds were used as a template.

We investigated the crystallization process, including spin coating, antisolvent dripping, and thermal annealing steps, using synchrotron-based in situ wide-angle x-ray scattering (WAXS) measurements to observe the phase evolution and structural dynamics. Briefly, the thin films were deposited by a spin-coating perovskite precursor solution of $\text{FAI}:\text{PbI}_2$:additive (in a 1:1.05: x mol % ratio) through a resistive-heating spin coater in the in situ measurement setup under nitrogen atmosphere; remote antisolvent dripping was performed with a remotely controlled pipette (fig. S4A). This in situ experiment comprised three steps. The coating step (spin coating and antisolvent dropping) mostly revealed the formation of the 2H-FAPI phase. The phase-evolution step (annealing from 25° to 150°C in steps of 10°C for 20 s) involved the formation of intermediate phases. The phase-stabilization step (annealing at 150°C continuously, for up to 500 s) demonstrated the formation of the 3C phase (fig. S4B). We investigated three types of samples: (i) chloride-free FAPI systems (a control film without any additive and FAPI-2D film with 2D BA_2PbI_4 additive), (ii) FAPI with chloride additives (MACl, FACl, and PbCl_2), and (iii) FAPI with coadditives (a combination of chloride additive such as FACl or MACl with 2D BA_2PbI_4 additive).

We first investigated the crystallization of bare FAPI without any additive (control) illustrated by the in situ contour WAXS plot (fig. S5A). The face-sharing hexagonal 2H phase of FAPI immediately formed at the beginning of the coating step, as indicated by the strong emergence of a $(100)^{2\text{H}}$ peak (at $q = 0.84 \text{ \AA}^{-1}$, where q is the length of the reciprocal lattice vector). A sudden transformation was observed at the end of the phase-evolution step from the 2H to corner-sharing 3C-FAPI phase [at $q = 0.99 \text{ \AA}^{-1}$, $(001)^{3\text{C}}$ peak] at ~130°C without forming any intermediate phase.

We also studied the effect of 2D seed (BA_2PbI_4) in the perovskite precursor solution (fig. S5B). At the beginning of the phase-evolution

step, a 2D intermediate phase appeared at $q = 0.65 \text{ \AA}^{-1}$ and $q = 0.96 \text{ \AA}^{-1}$ along with the 2H-FAPI phase and corresponded to the formation of $\text{BA}_2\text{FAPb}_2\text{I}_7$ ($n = 2$, BA-based 2D perovskite), as demonstrated by the out-of-plane $(400)^{2\text{D}}$ and $(600)^{2\text{D}}$ peaks, respectively. This 2D $n = 2$ intermediate provided a template for the growth of stable 3C-FAPI, and the transformation started at ~100°C, consistent with the previous results (6). Thus, in situ crystallization experiments on the chloride-free perovskite systems always revealed a direct transition from the 2H to the 3C phase of FAPI.

Next, we investigated the effect of MACl, FACl, and PbCl_2 chloride additives on the FAPI formation mechanism using in situ WAXS. First, we performed the same in situ experiment with 15 mol % MACl additive, which is the most commonly used method to obtain the black 3C-FAPI phase (fig. S6A). We observed the formation of 6H intermediate phases [$q = 0.88 \text{ \AA}^{-1}$, $(101)^{6\text{H}}$] along with 2H-FAPI during the phase-evolution step. The incorporation of MA and Cl led to a steplike growth pathway from the 2H through 6H to 3C phase and lowered the temperature of the 2H-to-3C lattice reconstruction down to 70°C. The film with 35% MACl revealed a much faster transition through a similar crystallization pathway because of its high chloride content (fig. S6B).

We investigated FACl in FAPI systems with concentrations of 5, 10, 15, and 20 mol % (fig. S7, A to D). For the 5 and 10 mol % FACl samples, two new peaks appeared at $q = 0.83 \text{ \AA}^{-1}$ and $q = 0.94 \text{ \AA}^{-1}$ during the 2H-to-3C lattice reconstruction. We attributed the peak at $q = 0.83 \text{ \AA}^{-1}$ to the (100) peak of the 8H phase, the 4H phase, or both. We attributed the peak at $q = 0.94 \text{ \AA}^{-1}$ to the (102) peak of the 8H phase, the (101) peak of the 4H phase, or both. The 8H polytype is the fingerprint of FAPI crystallization with FACl that completes the phase-evolution pathway, but it has rarely been discussed (27).

The formation of the 8H phase with FACl additive was further confirmed using 1D powder XRD measurements (fig. S8). However, the 4H and 8H diffraction peaks appeared within a similar 2θ range, making distinguishing between these two phases challenging. Thus, these features are hereafter denoted as 4H/8H. We note that the PbI_2 and $\text{PbI}_{2-x}\text{Cl}_x$ phases do not coincide with these diffraction features (fig. S9). The FAPI incorporated with 15 mol % FACl also revealed a $(101)^{6\text{H}}$ peak in addition to the 4H/8H peaks and provided a complete staircase lattice reconstruction from 2H-FAPI through 4H, 6H, and 8H to 3C-FAPI. The lattice reconstruction of FACl-stabilized FAPI starts at ~70°C. The PbCl_2 additive in the FAPI system also revealed the formation of 4H/8H polytype during film formation, but sluggish crystallization kinetics resulted in poor film quality (fig. S10). The FACl additive revealed a gradual stepwise phase-transition pathway through more corner-sharing octahedra (through the 4H, 6H, and 8H polytypes) among these three chloride additives. However, merely the presence of a chloride additive did not result in stable FAPI films.

We hypothesized that a coadditive synergistic strategy of using FACl and 2D perovskites could use Cl to control the crystallization pathway through the polytypes and the 2D phase to template and stabilize the FAPI black phase. As anticipated, FAPI formation kinetics with 15 mol % FACl and 0.5 mol % BA_2PbI_4 demonstrated a staircase lattice reconstruction from 2H-4H-6H-8H to 3C-FAPI that was similar to the formation pathway with 15 mol % FACl additives (Fig. 2A). The precise role of each additive will be discussed in the next sections, where we will show that the synergistic combination of FACl along with 2D BA $n = 1$ regulates the crystallization pathway to achieve a FAPI film with improved crystallinity and collectively induces a compressive strain that is required to stabilize the FAPI phase. The coadditive-treated film (a 15 mol % FACl and 0.5 mol % BA_2PbI_4 combination) is hereafter referred to as the FAPI-CA film.

We performed multiple experiments with different concentrations of FACl and 2D additives in FAPI. In all of the additive concentration combinations, the lattice-reconstruction process appeared to be mostly identical and revealed the formation of the 4H/8H phase during phase transition (figs. S11, A to C, and S12, A to C). Even at a high concentration

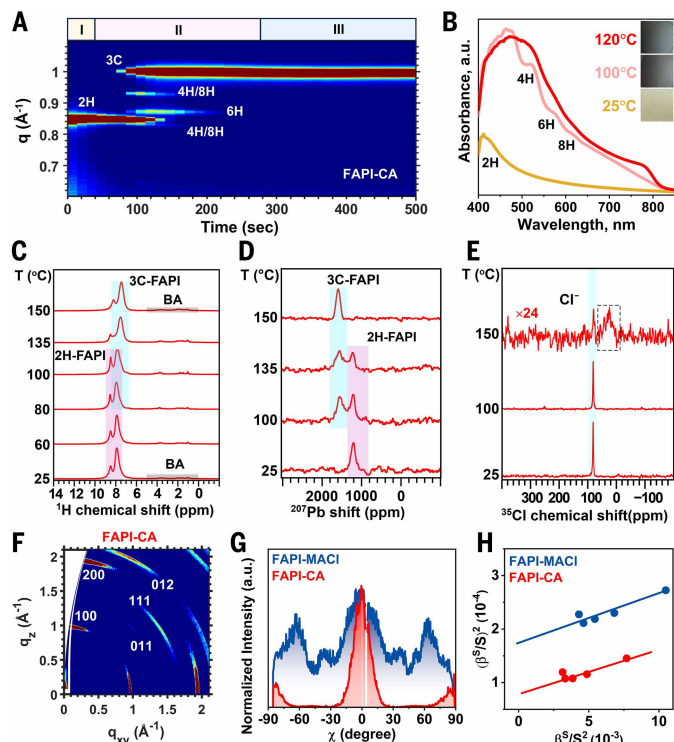


Fig. 2. In situ FAPI formation and structural characterization of the films. (A) In situ film formation with WAXS obtained from FAPbI₃ with 15 mol % FAPbI₃ additive in FAPbI₃ where the process comprised three steps: the coating step (I), phase-evolution step (II), and phase-stabilization step (III) shown at the top. (B) Absorption spectra of the FAPbI₃ film annealed at different temperatures. a.u., arbitrary units. (C) ¹H, (D) ²⁰⁷Pb, (E) ³⁵Cl ssNMR spectra of FAPbI₃ with 15 mol % FAPbI₃ and 2 mol % BA₂PbI₄ obtained as a function of annealing temperature, as indicated. (F) GIWAXS pattern of the FAPI-CA film. (G) The GIWAXS pattern integrated azimuthally along the (100) Debye-Scherrer ring of the FAPI-MAcI and FAPI-CA films, where the FAPI-CA films reveal highly oriented growth along the (100) direction. (H) Linear fit of integral-breadth analysis using Halder-Wagner plots, where β is the integral breadth of the diffraction peak and these integral breadths have been turned into scale of the variable $S = q/2\pi$, demonstrating that FAPI-CA films have lower microstrain and longer coherence length.

of 2D crystal (2 mol %), there was no signature of 2D peak formation, suggesting that the kinetics of the film formation was much faster. This result was in contrast with our previous study with 2D perovskites alone (6). Thereafter, we studied the FAPI system with 15 mol % MAcI and different concentrations (0.25, 0.5, and 1 mol %) of BA₂PbI₄ (fig. S13, A to C). In all cases, FAPI lattice reconstruction followed the pathway of 2H-6H to 3C, which was almost identical to the formation pathway with the addition of MAcI. All of the chloride-2D additive combinations showed a transition pathway similar to that of chloride additive alone, implying that the 2D additives did not alter the lattice-reconstruction pathway in the presence of chloride (all results are summarized in table S1). Notably, the FAPI-CA film revealed a systematic 2H-to-3C phase transition through the progressively increasing polytype staircase as intermediates among all of these additive combinations.

We performed an in situ WAXS measurement with conventional heating conditions rather than slow heat ramping, which replicated the process used in device fabrication. Similar to the slow heat ramping, the conventional heating likewise revealed the formation of polytypes during the FAPI-CA crystallization (fig. S14). Therefore, the slow temperature ramp did not influence the crystallization mechanism. Instead, it only elongated the 2H-FAPI-to-3C-FAPI phase transition through

the same polytypes and provided better insight into the intermediate phases.

We disentangled the respective roles of FA and Cl in the FAPI film formation by comparing the crystallization process with FAPbI₃, FAPbBr₃, and FAPbI₃ additives while keeping the 2D perovskite in the solution. We annealed the FAPI-CA (FAPI-2D-FAPbI₃), FAPI-2D-FAPbBr₃, and FAPI-2D-FAPbI₃ films at 60° and 80°C for several minutes and conducted in situ 1D XRD measurements (fig. S15, A and B). The FAPI-CA films crystallized through a sequential phase transition from 2H through 4H-6H-8H to 3C-FAPI, which was consistent with in situ WAXS measurements. By contrast, although the addition of FAPbBr₃ or FAPbI₃ lowered the 3C phase-transition temperature, the crystallization pathways of the FAPI-2D-FAPbBr₃ and FAPI-2D-FAPbI₃ films were similar to that of the control FAPI film (fig. S5A) without any additive, directly transforming from the 2H to 3C phase. For the crystallization of FAPI-2D-MAI, we did not observe the formation of the polytypes during its transformation to the black phase (fig. S15, C and D). To further elucidate the role of Cl and Br, we compared the FAPI crystallization process with that of the FAPI-BA₂PbCl₄ and FAPI-BA₂PbBr₄ films (fig. S16). The in situ XRD study revealed that the FAPI-BA₂PbCl₄ crystallization proceeds through the formation of different polytypes, whereas the FAPI-BA₂PbBr₄ crystallization demonstrates a direct conversion from the 2H to 3C phase. All of these experiments demonstrated the critical role of Cl-based additive in the FAPI formation process.

We recorded ultraviolet-visible (UV-vis) absorption spectra for the FAPI-CA films after 10 min of annealing at different temperatures (Fig. 2B). The film turned black at 100°C and showed several peaks in the absorption spectra, which we attributed to the presence of quantum-confined FAPI polytypes within the 3D bulk film (33). Finally, a complete conversion to 3C-FAPI occurred at 120°C, which is much lower than the additive-free FAPI phase-transition temperature. The FAPI-FAPbI₃ and FAPI-2D samples also revealed the formation of the 3C phase at 120°C (fig. S17). This is consistent with the in situ WAXS results, which show that the phase transition starts for the FAPI-CA and FAPI-FAPbI₃ at 70°C, whereas it begins for FAPI-2D at 100°C.

Time-resolved in situ solid-state NMR (ssNMR) spectroscopy revealed the local packing interactions during the phase transitions (34). These measurements were conducted under a high magnetic field of 28.2 T (¹H = 1200.5 MHz) to obtain high sensitivity and resolution. Figure 2, C to E, presents 1D ¹H, ²⁰⁷Pb, and ³⁵Cl magic-angle spinning NMR (MAS NMR) spectra of polycrystalline FAPI with 15 mol % FAPbI₃ and 2 mol % 2D (BA₂PbI₄) powders before and after thermal annealing at different temperatures (range of 25° to 150°C) as indicated. Specifically, ¹H and ²⁰⁷Pb NMR chemical shifts were sensitive to their local bonding environments in FA⁺ and [PbI₆]⁴⁻ octahedra, allowing us to resolve these moieties in the 3C and 2H phases. For example, at 25°C, two well-resolved ¹H signals in the 7 to 9 parts per million (ppm) range (FA⁺ in δ -phase) and a set of low-intensity peaks (gray box appears between 0 and 5 ppm in fig. 2C) associated with the BA⁺ cations were identified and distinguished. This finding was further corroborated with the ²⁰⁷Pb shift at ~1200 ppm for the edge-shared 2H-FAPI and ³⁵Cl NMR shifts of Cl additives. The materials annealed below 80°C exhibited identical ¹H features, but in the materials annealed between 80° and 100°C, FA⁺ cations coexisted in both the 2H and 3C-FAPI phases with different concentrations, and relatively narrower features were seen for the BA⁺ cations.

The in-between annealed material (100° to 135°C) exhibited coexisting signals consistent with polytype intermediates. For example, in the 135°C annealed materials, peaks with clearer ¹H and ²⁰⁷Pb NMR signatures of 3C-FAPI were detected, as evidenced by two different ²⁰⁷Pb shifts at ~1200 ppm (2H-FAPI) and 1600 ppm (3C-FAPI). Further annealing at 150°C stabilized 3C-FAPI. Formation of the black phase was supported by changes in the local environments of FA⁺ cations from the edge-shared nonperovskite phase into the corner-shared perovskite phase (cuboctahedron environment). The ¹H MAS NMR spectrum,

which initially showed signals corresponding to $-\text{CH}$ (8.5 ppm) and $-(\text{NH}_2)_2$ groups (7.6 ppm) of FA^+ cations in the yellow phase, displayed signals associated with $-\text{CH}$ (8.1 ppm) and $-(\text{NH}_2)_2$ groups (7.4 ppm) of FA^+ cations after thermal annealing and conversion into the black phase.

At the crystallographic level, several intermediate forms may exist (2H at 60°C and peaks for 4H, 6H, and 8H morphologies in between 80° and 150°C). However, the NMR signatures of the FA^+ cations were related to the local environment, surrounding either corner-shared or face-shared lead iodide octahedra, or combinations of them, which broadened the ^1H peaks. These results were corroborated by ^{207}Pb -NMR spectra, which displayed a single peak at 1200 ppm for the face-shared lead-iodide octahedra (2H-FAPI) in the pristine material and around 1600 ppm for the corner-shared lead iodide octahedra (3C-FAPI) for the 150°C annealed materials. The coexistence of these two phases, with different peak intensities, in the 100° to 135°C annealed materials, further supports the existence of different FAPI intermediate products (Fig. 1 A). In addition, ^{35}Cl ssNMR measurements and analysis confirmed the loss of chloride, because the peak intensity of ionic-like chloride species (80 ppm) decreased upon annealing between 100° and 150°C. A broad peak in the vicinity of 0 to 100 ppm appeared that we attributed to the formation of minuscule concentrations of Cl alloys in the perovskite phase. Although it is difficult to quantify its concentration, this peak provides direct evidence of chloride doping into the final 3C-FAPI lattice.

For the FAPbI₃-only additive (15 mol %), ^1H , ^{207}Pb , and ^{35}Cl NMR confirmed stabilization of the black phase through a slightly different pathway for full 2H-to-3C reconstruction (fig. S18, A to C). Thermal annealing from above room temperature to 100°C produced identical peak patterns. Lattice reconstruction occurred at 135°C, as evidenced by the broadening of the peaks (plausibly arising from a minor concentration of δ -FAPbI₃) or line-shape distortions in ^1H and ^{207}Pb spectra; a full 3C transformation was achieved by 150°C. Chloride incorporation again manifested as a diminishing 80-ppm peak and broad features around 0 to 100 ppm.

In contrast to the chloride additives, 2D Ba_2PbI_4 seeds induced a different mechanism in the conversion and stabilization of 2H-FAPbI₃ to 3C-FAPbI₃ (fig. S19, A and B). In this case, PbI_2 was evident in the ^{207}Pb spectrum of the fresh material at room temperature, which reduced its intensity at 100°C and disappeared at 150°C, suggesting that the conversion mechanism was mediated through the consumption of PbI_2 . A substantial fraction of black phase was already formed (also evidenced by the consumption of PbI_2) after annealing at 100°C. The concentration of 3C-FAPbI₃ surpassed 2H-FAPbI₃ in both ^{207}Pb and ^1H ssNMR spectra, which was estimated based on the peak intensities. These results indicate that the phase transition was more uniform with FAPbI₃-2D coadditive systems than with a single additive.

For the state-of-the-art FAPI system with MACl additive (35 mol %), the additional ^1H signals we observed arose from the presence of MAPbI_3 and MAI/MACl moieties, as well as the different distributions of FA^+ cations with plausible existence of $\text{FAPbI}_3/\text{FAI}$ species (fig. S20A). Lattice reconstruction could be tracked through the changes in the ^1H NMR peaks of MA and FA. At 25°C, the peaks exhibited different intensities compared with those of the annealed materials at 120°C. After annealing at 150°C, well-resolved FA^+ peaks corresponding to 3C signals were present, although the peaks corresponding to MA^+ persisted, indicating the presence of the MAPbI_3 phase and consistent with previous results (35). This result was consistent with the analysis of ^{207}Pb NMR spectra that showed peaks corresponding to the presence of both FAPbI_3 and MAPbI_3 phases, and a solid solution of these two cations, for example, $\text{FA}_x\text{MA}_{1-x}\text{PbI}_3$ and by-products such as FAPbI_3 and MACl (fig. S20B). The ^{35}Cl ssNMR data of FAPbI_3 -MACl revealed the persistence of chlorine signals after annealing at 150°C, similar to the FAPbI_3 -2D-FAPbI₃ sample in Fig. 2C (see also fig. S20C). Although Cl volatilization was more favorable with MACl than FAPbI₃ because it more readily deprotonated (36), the ^{35}Cl ssNMR spectral patterns

showed more similarities than differences. The observed ssNMR signal related to Cl in the lattice for FAPbI_3 -MACl, even after rapid volatilization, likely arose from the high concentration of additive (35 mol %).

We estimated the excess FA in the FAPI sample from the FA peak integrals that were calibrated against the ^1H peaks of BA, taken as unity. The FA concentration remained comparable for both the FAPbI_3 -2D and FAPbI_3 -2D-FAPbI₃ samples annealed at 150°C. We propose that excess FA may be removed from the sample during the antisolvent spin-coating or washing step and further removal could proceed as FAPbI_3 can decompose through *sym*-triazine formation (37, 38). The removal of excess FA can be further confirmed from the time-of-flight secondary ion mass spectrometry (ToF-SIMS) depth profile of the FAPI-2D and FAPI-CA films demonstrating similar FA content (fig. S21). These investigations were consistent with the in situ WAXS analysis, which established that the chloride additives unlock a crystallization pathway through the polytypes that only formed in the presence of chloride, and without which was not observed.

All of these in situ formation results showed that the FAPI formation with 2D-FAPbI₃ coadditives provided a superior crystallization pathway and highly crystalline films through these polytypes, enabled by the incorporation of chloride into the 3C-FAPbI₃ lattice. In previous reports, the Cl-based additives into perovskite precursor solution only regulated the crystallization, and the chloride left the perovskite films as HCl gas (15). Some reports have suggested that the residual chloride passivated grain boundaries of the FAPI film, which eventually has a detrimental effect on device stability (39). Even after using a high additive concentration, reports of the incorporation of chloride in perovskite lattice are rare. Only a few studies have shown chloride incorporated into the perovskite lattice, but these reports have been restricted to MAPbI₃ (or MA-doped perovskite) or the wide-bandgap system where Br-Cl alloys are formed (40, 41).

After probing film formation, we characterized the MACl and coadditive-treated final FAPI films with grazing incidence WAXS (GIWAXS) measurements to gain insight into the film crystallinity. The film with MACl additive revealed a continuous Debye-Scherrer ring, suggesting a random orientation of grains and high mosaicity (fig. S22). The azimuth angle mapping, derived from the perovskite (100) plane during the in situ WAXS measurements (fig. S23A), displayed a broad distribution that was quite consistent with the resultant film. By contrast, the film formation with 2D-FAPbI₃ coadditives revealed a sharp peak in the in situ azimuth angle mapping, signifying a nucleation stage with a favored out-of-plane (100) facet that resulted in a highly oriented polycrystalline film (fig. S23B). The FAPI-CA film likewise exhibited a well-defined Bragg diffraction spot along the (100) plane (Fig. 2F), which was further evident from the azimuthal-dependent intensities extracted through integrating 1D intensity data for the α -FAPbI₃ (100) plane in the GIWAXS pattern (Fig. 2G).

The preferred azimuthal angle at 0° of the FAPI-CA film suggested a strong (100) orientation, whereas the MACl-treated film demonstrated a mixed orientation. This observation implied reduced mosaicity and a higher degree of perovskite orientation in the out-of-plane direction for the coadditive-treated film. The results with individual additive demonstrated that the 2D additive helped in orienting the coadditive-treated film (fig. S24). Moreover, Halder-Wagner analysis revealed a lower y intercept and slope for the FAPI-CA film, suggesting minimized microstrain and a high coherence length, respectively (see supplementary methods and Fig. 2H).

We conducted scanning electron microscopy (SEM) and atomic force microscopy (AFM) analyses to gain insight into the film morphology. The SEM image of the FAPI-MACl film revealed small and irregular grains of perovskite (fig. S25A), but the FAPI-CA film showed continuous and smooth perovskite films with large grains resulting from controlled crystallization. AFM analysis revealed large-grain FAPI-CA films with lower roughness (fig. S25B). Optical characterization of the FAPI-MACl and FAPI-CA films with UV-vis absorption spectroscopy revealed that

FAP-MAcI film exhibited a bandgap of 1.53 eV, whereas the FAP-CA film demonstrated a slightly lower value of 1.50 eV (fig. S26A). We attributed this bandgap reduction to the lower lattice disorder and the exclusion of MA. Time-resolved photoluminescence showed that the FAP-CA film displayed a slower photoluminescence decay rate than the FAP-MAcI sample (fig. S26B), consistent with fewer defect states leading to superior charge-carrier dynamics and reduced nonradiative recombination. These *in situ* formation and film characterizations indicate that the FAP-CA film exhibited a superior crystallization pathway through all the polytypes, resulting in high-quality Cl-doped FAP film.

Degradation mechanism of FAP-CA films

The measurements reported in the previous section elucidate the correlation between how different additives result in a different formation pathway. To fundamentally understand the correlation between the formation pathways and the degradation of the FAP films, we performed accelerated aging measurements. We first performed aging under 1-sun illumination and 90°C by monitoring the XRD peaks from the different additive-stabilized FAP films. Although the control MAcI-treated FAP film degraded quickly, the FAP-CA film did not undergo any sort of decomposition after 1000 hours (fig. S27).

To induce degradation, we performed accelerated aging of the perovskite films under 15-sun illumination and inert atmosphere while progressively increasing the temperature from 65° to 90°C (Fig. 3A). During this process, the temperature was increased in three steps: The films were first held at 65°C for 175 hours, followed by 75°C for the next 125 hours, and then at 90°C thereafter. FAP-MAcI films showed the fastest rate of degradation among all the films studied. The XRD (100) and (200) peaks of the FAP-MAcI samples corresponding to the 3C-FAP phase gradually vanished after 225 hours, most likely because of the disruption of the 3C-FAP lattice triggered by the volatilization of MA. The presence of substantial amounts of MA was evidenced by the ssNMR analysis and was consistent with previous studies (35). After 250 hours of illumination, the film turned yellow, and both the (100)

and (200) XRD peaks completely lost all their intensity. Interestingly, a new peak at 28.3° arose that was slightly shifted from the (200) peak along with a small peak at 12.67°. The XRD pattern of this new degradation product was stable even after 400 hours of illumination, and no signature of 2H-PbI₂ phase (hexagonal PbI₂) formation was observed, suggesting that the MAcI-stabilized FAP films formed a degradation product not previously reported.

The FAP-2D film (no chloride additive) was stable up to 225 hours, after which point a broadening of 3C-FAP peaks was observed followed by a degradation through 2H-FAP and then 2H-PbI₂. The 2H-FAP phase reduced over time, and, finally, only 2H-PbI₂ remained after 400 hours. The 2D-stabilized FAP thus underwent degradation through 2H-FAP to 2H-PbI₂.

In contrast to the above films, the FAP-CA film was extremely stable up to 350 hours and finally degraded to a product similar to that of the FAP-MAcI film, bypassing the formation of 2H-FAP. We propose the following possibilities for the new degradation product: (i) a polytype of FAP or (ii) a different phase of PbI₂. However, the XRD pattern of this new degradation product did not match that of any FAP polytype but rather matched that of the previously reported 3R phase of PbI₂ [Cambridge Crystallographic Data Centre (CCDC) deposition no. 1600311] (42). The 3R-PbI₂ polytype exhibits multiple interplanar spacings that are analogous to those of cubic FAP, indicating considerable structural similarity (43). The XRD patterns presented in Fig. 3B suggested that the FAP-CA film was slightly compressed by chloride incorporation and exhibited a d-spacing closer to the (104) plane of 3R-PbI₂, thereby facilitating the thermodynamically driven transition. It is worth noting that the compressive strain associated with the Cl incorporation is different from that introduced with 2D addition. The compressive strain because of the $n = 2$ 2D Ruddlesden-Popper phase that we observed in our previous work is due to both an in-plane and out-of-plane (lateral) templating effect between the 3D domains and the 2D nanodomains (6, 13). Although this is an effective strategy, it does not completely overcome the nucleation of the yellow phase inside the 3D nanodomains with accelerated aging (Fig. 3A). As a result, the mesoscopic effect described previously is still expected in this present work with the 2D perovskite phase we are using, but in addition, we propose that the dispersed Cl atoms at the perovskite octahedra level are able to induce a local compressive strain (understood here as a bond-shortening effect), which is necessary for progressively stepping through the increasingly corner-sharing polytypes, thereby quenching the yellow-phase nucleation. We also characterized the degraded films with UV-vis absorbance and GIWAXS. All three degraded FAP films showed a similar absorbance edge arising from PbI₂ formation, indicating that their electronic properties may be similar (fig. S28). The GIWAXS pattern of the degraded FAP-CA film revealed a highly oriented 3R-PbI₂ phase that was perfectly aligned with the (200) peak of the 3C phase in the initial film in Fig. 2F (see also Fig. 3C).

The FAP-CA films had a much higher threshold for degradation, thus presenting the best-case scenario for realizing the intrinsically most stable perovskite film (Fig. 3D). To further confirm the degradation, we performed accelerated thermal stability on the FAP-CA, FAP-2D-FABr, and FAP-2D-FAI films where we heated the sample at different temperatures under inert atmosphere without illumination. The FAP-CA film began to degrade at 250°C, demonstrating a coexistence of 3C-FAP and 3R-PbI₂, eventually completely converting to a 3R-PbI₂ phase at 300°C (fig. S29). By contrast, the FAP-2D-FABr, and FAP-2D-FAI films readily degraded to 2H-PbI₂ after 10 min of annealing at 250°C (fig. S30). Thus, lattice Cl improved FAP stability by providing a different degradation pathway. These results also show that 3R-PbI₂ is the thermodynamically stable phase and the activation energy is much higher than that for the 2H-PbI₂ phase. Therefore, in this study, the Cl additives enabled a stepwise transition pathway, and both the 2D and Cl additives collectively induced a compressive strain that provides excellent stability.

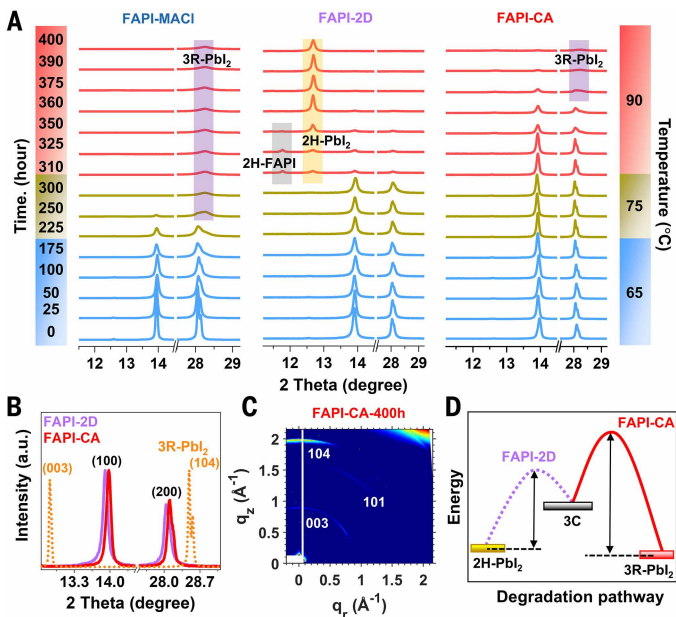


Fig. 3. Degradation mechanism of the FAP film. (A) Degradation of MAcI, 2D, and coadditive-treated FAP films under 15-sun illumination at temperatures of 65°, 75°, and 90°C with 1D XRD measurements. (B) XRD pattern of FAP-2D, FAP-CA, and 3R-PbI₂. (C) GIWAXS image of FAP-CA-based FAP films after 400 hours of aging, which demonstrates the formation of 3R-PbI₂. (D) Schematic illustration of the degradation pathway of 3C-FAP fabricated with the 2D and coadditive methods.

Photovoltaic devices and stability

To evaluate the notable durability of the FAPI-CA films in a device stack with electron- and hole-transport layers, we fabricated p-i-n devices with an architecture of FTO/Mix-PACz/FAPI/C₆₀/BCP/Cu (Fig. 4A) and a 0.5-cm² active area (FTO, fluorine-doped tin oxide; Mix-PACz, mixture of MeO-2PACz and Me-4PACz in 2:3 ratio; BCP, bathocuproine). The details of the device fabrication are discussed in the supplementary materials and follow a standard process used to fabricate such devices. The FAPI-CA device was optimized with different concentrations of FAPbI₃ and 2D additives in the FAPI precursor solution. The best device performance was obtained for 15 mol % FAPbI₃ and 0.5 mol % 2D additive concentrations. The current density–voltage (*J*-*V*) characteristics in a forward and reverse scan of the champion device are presented in Fig. 4B and show minimal hysteresis. Under standard air mass 1.5 global (AM1.5G) illumination, the device exhibited a PCE of 25.1%, with a short-circuit current density (*J*_{SC}) of 26.4 mA cm⁻², a *V*_{OC} of 1.15 V, and a fill factor of 0.82. The FAPI-MAcI device exhibited a relatively lower PCE of 23.6% than the FAPI-CA device because of inferior film quality and higher recombination (fig. S31A). Figure 4C and fig. S31B illustrate the external quantum efficiency (EQE) of the champion devices. The integrated *J*_{SC} estimated from the EQE is in good agreement with the *J*-*V* measurement. Furthermore, the repeatability of the FAPI-MAcI and FAPI-CA devices fabricated in different batches is presented in box plots (fig. S32).

Finally, we investigated the effect of the coadditive strategy on the long-term operational stability of the FAPI device and compared that with state-of-the-art MAcI additive. For the stability measurements, we replaced the organic BCP layer with an inorganic SnO₂ layer deposited using atomic layer deposition in the device. Initially, we performed

the stability test on the encapsulated FAPI-CA device at MPPT under 1-sun illumination and 85°C in ambient air of 60 to 70% relative humidity [International Summit on Organic Photovoltaic Stability (ISOS) ISOS-L-2 protocol]. The FAPI-CA device demonstrated excellent stability and retained >99% of its initial PCE after 500 hours (fig. S33A). Furthermore, we aged the encapsulated devices in a commercially sourced aging box (1-sun illumination without any UV filter) under an open-circuit condition in ambient air with a relative humidity of 60 to 70% (ISOS-L-1 protocol). We used an open-circuit condition rather than MPPT because an accumulation of photogenerated carriers occurs under open-circuit conditions (44, 45). At different time intervals, the FAPI-CA devices were removed from the aging environment and characterized under AM1.5G illumination (Fig. 4D). The devices fabricated with the coadditive approach showed no degradation, with a *T*₉₉ exceeding 1000 hours (10 cells). By contrast, the FAPI-MAcI devices degraded to 50% of their initial efficiency after 200 hours. Because negligible degradation was observed for the FAPI-CA devices, they were subjected to an even higher thermal stress of 85 ± 5°C under an open-circuit condition with a relative humidity of 60 to 70%. Under *V*_{OC} conditions and 85 ± 5°C, the FAPI-MAcI device degraded readily, whereas the FAPI-2D device retained 89% of its initial PCE after 600 hours of continuous illumination (fig. S33B). The FAPI-CA device showed almost no degradation and retained 99% of its initial PCE after 600 hours. After 1200 hours of continuous testing, the device retained >98% of its original PCE, which was even better than results obtained on films with ~50-nm poly(methyl methacrylate) (PMMA) coating (Fig. 4E).

Discussion

We attribute the excellent durability of FAPI-CA devices to bypassing the 2H-PbI₂ phase and instead inducing degradation via the energetically uphill 3R phase of PbI₂, which is consistent with the results on thin films as shown in Fig. 3 and with theoretical prediction. These results represent state-of-the-art durability in a single-junction perovskite solar cell and pave the way for the investigation of formation and degradation pathways and their critical role in obtaining commercially relevant durability. We anticipate that bypassing specific degradation pathways may be important for other compositions with mixed-halide and tin-based perovskites.

Materials and methods are available in the supplementary materials.

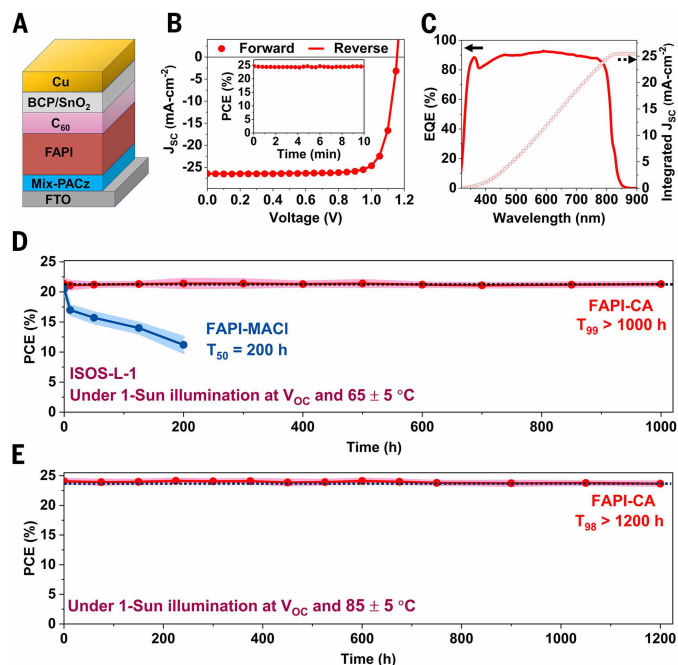


Fig. 4. Photovoltaic performance and stability under open-circuit conditions.

(A) Schematic of the device architecture. (B) The *J*-*V* characteristics of the champion FAPI-CA device and the stabilized output of the champion device tracked at the maximum power point under standard AM1.5G illumination (inset). (C) EQE and integrated *J*_{SC} of the champion device. (D) Long-term stability of the encapsulated MAcI and coadditive-treated FAPI devices (10 devices for both) aged under *V*_{OC} conditions and continuous 1-sun illumination (65° ± 5°C) in ambient air with 60 to 70% relative humidity. (E) Accelerated aging on the encapsulated coadditive-treated device under *V*_{OC} conditions and continuous 1-sun illumination at 85° ± 5°C in ambient air (60 to 70% relative humidity).

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S33; Table S1; References (47–55); Movie S1

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Bypassing the yellow phase for extremely stable formamidinium lead iodide perovskite solar cells

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Editor's summary

Two additives cause the chloride ion to incorporate into formamidinium lead iodide perovskites, which limits degradation into the photoinactive yellow phase. Garai *et al.* showed that adding formamidinium chloride and Bu₂PbI₄ (where Bu is butylammonium) created compressive lattice strain that stabilized the photoactive black phase and made the formation of the yellow phase during degradation unfavorable. Solar cells with an average power conversion efficiency of 24.1% had only a 2% loss after 1200 hours at 90°C and open circuit conditions. —Phil Szuromi

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